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Distributed online optimal power flow for distribution system

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ABSTRACT

This paper proposes a distributed online OPF algorithm for distribution systems with high integration of electronic-interfaced distributed energy resources (DERs). The algorithm can coordinate the update of DER set-points on fast time scales only depending on local measurements and boundary information for each distribution area. Comparing with the existing distributed online algorithms that require some special setups (e.g., without considering the constraints on currents), the proposed algorithm can cope with various setups for the objective functions and constraints. The algorithm is based on solving the Taylor approximations of OPF problems by a distributed solution. In this solution, we design a strategy to formulate the second-order Taylor information in a distributed way by decomposing the sensitivities of the OPF targets among different areas; moreover, we design a method to simplify the constraints on DER outputs to accelerate the solution. The validity of the algorithm is verified by case studies, which show that it can continuously drive DER outputs towards the time-varying OPF targets composed of power losses and operation costs of DERs while satisfying the constraints on voltages and currents.

1. Introduction

Future distribution systems are expected to integrate large numbers of electronic-interfaced distributed energy resources (DERs), such as photovoltaics (PVs) and battery energy storage systems (BESSs) [1–3]. They introduce fast control capacities, and provide the potential to coordinate the operation of distribution systems more accurately. However, most of the optimal power flow (OPF) algorithms, such as the classical reduced gradient method [4] and the Newton method [5] and some distributed OPF algorithms [6–9], are offline. The offline OPF algorithms require to iterate on all variables several times in cyberspace until getting an optimal solution that can be applied to the electrical systems. This solving process is time-consuming and unable to fully utilize the fast control capacities of DERs, as well as to respond to fluctuations of outputs of renewable energy sources (RESs) [10]. Future distribution systems require online OPF algorithms working on fast timescales.

Some recent efforts have focused on the online OPF algorithms. The online algorithms utilize the measurements to iterate only on the controllable variables, and simultaneously apply the intermediate iterates to update DER set-points and drive DER outputs towards given OPF targets in feedback interaction with the electrical systems [10]. Ref. [11] proposed a gradient-based online OPF algorithm for radial distribution grids and established sufficient conditions for its convergence to a global optimum. Ref. [12] proposed an OPF seeking controller for grids with arbitrary structure, and introduced semi-definite programming relaxations to achieve global convergence. Ref. [13] designed an adaptive feedback controller introducing a modeling approach in terms of manifold optimization. Ref. [14] proposed a model-free online OPF algorithm based on extremum seeking methods, where the gradients are estimated in procedures of injecting sinusoidal probing signals into electrical grids. Ref. [15] designed an OPF seeking controller based on alternating direction method of multipliers. Ref. [16] proposed an OPF-pursuit controller and analytically established its OPF-target tracking capabilities in terms of operation changes and measurement errors. Ref. [17] proposed a quasi-Newton-based online OPF algorithm, where the solution is accelerated by adaptively shrinking the size of Jacobian matrixes according to the binding constraints in real time. Ref. [18] proposed an incentive-based solution to track time-varying targets of maximizing social-welfare. Ref. [19] designed a gradient-based controller and studied its convergence when the set-points of DERs are discrete. Ref. [20] designed a stochastic dual algorithm for solving online OPF problems considering the multi-period constraints. However, the above online solutions [11–20] are all based on a centralized framework [10], where a center is needed to collect all the parameter and measurement data throughout the considered system

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to formulate the sensitivities related to given OPF targets. This framework faces the problem of single point of failure and therefore 'limited reliability' due to centralized data collections [21–24].

To overcome the shortage, some works [25–34] proposed the distributed online algorithm only depending on the local and/or neighbors' information. Refs. [25–31] deduced the sensitivities used to update DER set-points into a distributed fashion, and thus developed the online algorithm into a distributed solution; however, the deductions of these methods [25–31] were based on OPF models without considering the current constraints, and moreover, the consideration of power losses in [27,28,31] needs an assumption of homogeneity of the X/R ratios across the considered distribution grids, and [29,30] can not consider the hard constraints on voltages. Refs. [32,33] proposed the online algorithms for frequency control, but they were based on DC power flow models without considering voltage regulation and power losses minimization. Ref. [34] proposed a heuristic-based algorithm for minimizing power losses, but the consideration of the other objectives (e.g., minimization of the operation costs of DERs) needs further improvements for the heuristic strategy.

To this end, this paper proposes a novel distributed online OPF algorithm. Comparisons between the proposed algorithm and the others [25–34] are summarized in Table 1. One can see that the proposed algorithm has more extensive application scopes and can cope with various setups for the objective functions and constraints. The algorithm is based on solving the Taylor approximations of OPF problems by a distributed solution. In this solution, we design a strategy to formulate the second-order Taylor information in a distributed way by decomposing the sensitivities of the OPF targets among different areas; moreover, we design a method to simplify the constraints on DER outputs to accelerate the solution. Case studies show that it can continuously drive DER outputs towards time-varying OPF targets while satisfying the constraints on voltages and currents.

Overall, the main contribution of this paper is three-fold:

(1) We proposed a distributed online OPF algorithm only depending on local measurements and boundary information for each distribution area. Comparing with the existing distributed online solutions, the proposed algorithm has more extensive application scopes and can cope with various setups for the objective functions and constraints.
(2) We design a strategy to formulate the Jacobian and Hessian of OPF problems for distribution systems in a distributed way. This is useful for the optimization and analysis of distribution systems where the information is unable to be processed centrally.
(3) We design a strategy to simplify the coupling constraints on DER outputs. It can accelerate the solution and is significant for achieving close-followed OPF tracking capacities.

Table 1
Comparisons of the proposed algorithm with the existing distributed online algorithms.

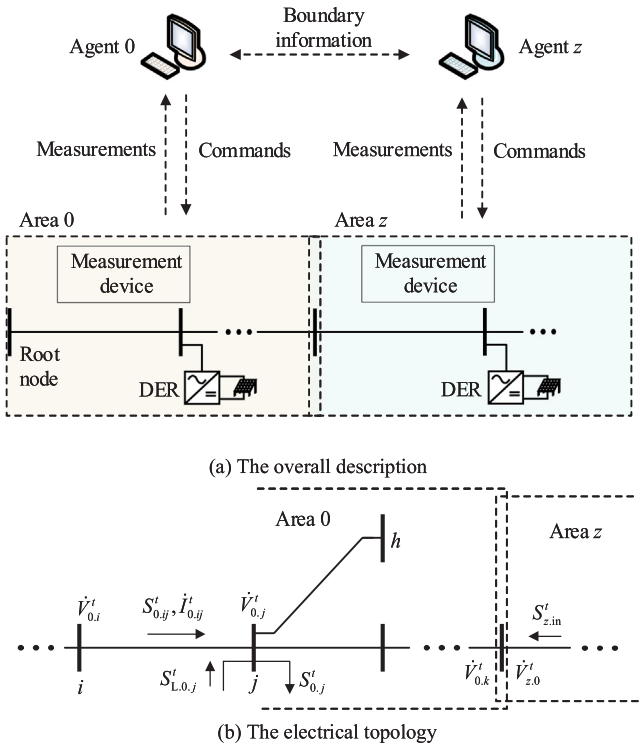
Method	Objective		Constraint			AC power flow (Yes/No)
	Power loss	Cost of generators	Output of generators	Voltage	Current	
[25]	√	×	×	√	×	Y
[26]	√	×	√	√	×	Y
[27 28]	√	×	√	√	×	Y
[29]	×	√	√	×	×	Y
[30]	×	×	√	×	×	Y
[31]	√	√	√	√	×	Y
[32]	×	√	√	×	×	N
[33]	×	√	√	×	√	N
[34]	√	×	√	√	√	Y
This paper	√	√	√	√	√	Y

Table 2
List of some notations.

Indexes and sets	
$[\bullet]^t_{z.ij}$, $[\bullet]^t_{z.i}$	Variables or parameters corresponding to line (i,j) and node i in area z at time instant t
$\mathcal{Z}^+$	Set of areas excluding the upper area
$\mathcal{Z}$	Set of areas; $\mathcal{Z} = \{0\} \cup \mathcal{Z}^+$
$\mathcal{N}_z^+$	Set of nodes excluding node 0 in area z; $\mathcal{N}_z^+ = \{1,...,N_z\}$ with N_z as the number of nodes in area z
$\mathcal{N}_z$	Set of nodes in area z; $\mathcal{N}_z = \{0\} \cup \mathcal{N}_z^+$
$\mathcal{M}_z$	Set of lines in area z; $\mathcal{M}_z \subset \mathcal{N}_z \times \mathcal{N}_z$
$\mathcal{B}$	Set of boundary node pairs relating area 0 and the other areas
$\mathcal{H}_{z.i}$	Set of nodes at the downstream of node i and adjacent to node i
$\mathcal{T}$	Set of time instants
$\Omega_{z.i}$	Set of DER set-points limited by their physical properties
Parameters:	
Δt	Time interval for the update of DER set-points
$\bar{V}$, $\underline{V}$	Upper and lower limits on voltages respectively
$\bar{I}_{z.ij}$	Upper limit on currents
$P^t_{L.z.i}$, $Q^t_{L.z.i}$	Active and reactive load power
$\bar{\boldsymbol{U}}_z$, $\underline{\boldsymbol{U}}_z$	$\bar{\boldsymbol{U}}_z = \bar{V}^2 \cdot \boldsymbol{e}^{N_z}$ and $\underline{\boldsymbol{U}} = \underline{V}^2 \cdot \boldsymbol{e}^{N_z}$ wtih $\boldsymbol{e}^{N_z}$ denoting the unit vector with N_z elements
$\bar{\boldsymbol{l}}_z$	$\bar{\boldsymbol{l}}_z = [...,\bar{I}^2_{z.ij}, ...]^{\mathrm{T}}$, $ij \in \mathcal{M}_z$
$z_{z.ij}$	Impedance of line (i,j); $z_{z.ij} = r_{z.ij} + jx_{z.ij}$ with $r_{z.ij}$ and $x_{z.ij}$ as resistance and reactance respectively
$\bar{\boldsymbol{\gamma}}_z$, $\underline{\boldsymbol{\gamma}}_z$	$\bar{\gamma}_z = [...,\bar{\gamma}_{z.i}, ...]^{\mathrm{T}}$, $\underline{\gamma}_z = \left[...,\underline{\gamma}_{z.i}, ...\right]^{\mathrm{T}}$, $i \in \mathrm{N}_z^+$ with $\bar{\gamma}_{z.i}$ and $\underline{\gamma}_{z.i}$ as the penalty coefficient corresponding to the upper and lower limits on voltages
$\boldsymbol{\gamma}_z$	$\gamma_z = [...,\gamma_{z.ij}, ...]^{\mathrm{T}}$, $i \in \mathrm{M}_z$ with $\gamma_{z.ij}$ as the penalty coefficient corresponding to the limit on currents
Variables:	
P^t_{loss}	Power losses of the considered distribution system
f^t_z	Time-varying objective function of area z
$P^t_{z.ij}$,$Q^t_{z.ij}$	*Sending-end* active and reactive power from node i to node j
$S^t_{z.ij}$	*Sending-end* complex power; $S^t_{z.ij} = P^t_{z.ij} + jQ^t_{z.ij}$
$P^t_{z.i}$, $Q^t_{z.i}$	Set-points of active and reactive power of DERs
$S^t_{z.i}$	Set-points of complex power of DERs; $S^t_{z.i} = P^t_{z.i} + jQ^t_{z.i}$
$P^t_{z.in}$, $Q^t_{z.in}$	Active and reactive power injected into area 0 from area z
$S^t_{z.\mathrm{in}}$	Complex power injected into area 0; $S^t_{z.\mathrm{in}} = P^t_{z.\mathrm{in}} + jQ^t_{z.\mathrm{in}}$
$\dot{V}^t_{z.j}$, $V^t_{z.i}$	Complex voltage and voltage magnitude
$\dot{I}^t_{z.ij}$, $I^t_{z.ij}$	Complex current and current magnitude
$\widetilde{P}^t_{z.ij}$,$\widetilde{Q}^t_{z.ij}$, $\widetilde{I}^t_{z.ij}$, $\widetilde{U}^t_{z.i}$	Estimates of $P^t_{z.ij}$,$Q^t_{z.ij}$, $I^t_{z.ij}$, $U^t_{z.i}$ respectively. This paper set the estimates at t to the measurements at $t-1$
$\boldsymbol{u}^t_{z.i}$	$\boldsymbol{u}^t_{z.i} = [P^t_{z.i}, Q^t_{z.i}]^{\mathrm{T}}$
$\boldsymbol{u}^t_z$	$\boldsymbol{u}^t_z = [...,\boldsymbol{u}^{t\mathrm{T}}_{z.i},...]^{\mathrm{T}}$, $i \in \mathcal{N}_z^+$
$\boldsymbol{U}^t_z$	$\boldsymbol{U}^t_z = [...,U^t_{z.i}, ...]^{\mathrm{T}}$, $U^t_{z.i} = V^{t2}_{z.i}$, $i \in \mathcal{N}_z^+$
$\boldsymbol{l}^t_z$	$\boldsymbol{l}^t_z = [...,l^t_{z.ij}, ...]^{\mathrm{T}}$, $l^t_{z.ij} = I^{t2}_{z.ij}$, $ij \in \mathcal{M}_z$
$\widetilde{\boldsymbol{U}}^t_z$, $\widetilde{\boldsymbol{l}}^t_z$	$\widetilde{\boldsymbol{U}}^t_z = [...,\widetilde{U}^t_{z.i}, ...]^{\mathrm{T}}$, $i \in \mathcal{N}_z^+$; $\widetilde{\boldsymbol{l}}^t_z = [...,\widetilde{l}^t_{z.ij}, ...]^{\mathrm{T}}$, $ij \in \mathcal{M}_z$
$\boldsymbol{g}^t_z$, $\boldsymbol{B}^t_{sz}$	Jacobian matrix and Hessian matrix
$\tau^{t.k}_{z.i}$	Calculation time for area z at the ith fragment of the kth iteration in $[t-\Delta t, t]$
τ^t	Minimum time for the calculation in $[t-\Delta t, t]$
diag($\boldsymbol{X}$)	the main diagonal whose row number is equal to the element number of the column vector $\boldsymbol{X}$, and the diagonal elements are equal to the elements of $\boldsymbol{X}$ at the same rows

Note that all the matrix differentiations in this paper are under the denominator-layout.

The rest of this paper is organized as follows. Section 2 describes a time-varying OPF problem for multi-area distribution systems. Section 3 presents the distributed algorithm to solve the OPF problem. Section 4 presents the case studies and the test results. Section 5 concludes this paper. The main symbols used in the paper and their meanings are listed in Table 2.



(a) The overall description

(b) The electrical topology

Fig. 1. Schematic representation of a multi-area distribution system.

2. Problem formulation

Consider a distribution system illustrated in Fig. 1. It is radial and contains several areas that include an upper area containing the root node and several lower areas connected to the upper area. Each area integrates various types of DERs, such as photovoltaics (PVs) and battery energy storage systems (BESSs). Moreover, each area is equipped with measurement devices, communication systems and calculation agents, by which each area can obtain the internal power flow states (may be handled based on state estimation methods [35,36]) in real time, as well as send control commands to the internal DERs. The measurements include the voltage magnitudes at each node, and the current magnitudes and the active and reactive flows on each branch. The internal operation details for each area, such as measurement and parameter data, are available to itself and private to each other, and only limited boundary information can be exchanged among different areas.

The purpose is how to coordinate the operation of all the areas to continuously drive their operation towards time-varying OPF targets. For this purpose, consider a time-varying OPF problem to optimize the operation of distribution systems:

$$\mathrm{Min} \kappa P_{\mathrm{loss}}^{t} + \sum_{z\in\mathcal{Z}} f_z^t \tag{1a}$$

s.t.

$$\begin{cases} \dot{V}_{z.j}^{t} = \dot{V}_{z.i}^{t} - z_{z.ij}\dot{I}_{z.ij}^{t},\ \dot{I}_{z.ij}^{t} = S_{z.ij}^{t*}/\dot{V}_{z.i}^{t*} \\ S_{z.ij}^{t} = \sum_{h\in\mathcal{H}_{z.j}} S_{z.jh}^{t} + S_{\mathrm{L}.z.j}^{t} - S_{z.j}^{t} + (\dot{I}_{z.ij}^{t*}\dot{I}_{z.ij}^{t})z_{z.ij} \\ \forall\, i\in\mathcal{N}_z,\ ij\in\mathcal{M}_z,\ z\in\mathcal{Z} \end{cases} \tag{1b}$$

$$\dot{V}_{0.i}^{t} = \dot{V}_{z.0}^{t},\ \forall\ (0.i,\ z.0)\in\mathcal{B} \tag{1c}$$

$$\begin{cases} \underline{V} \leqslant V_{z.i}^{t} \leqslant \bar{V} \\ I_{z.ij}^{t} \leqslant \bar{I}_{z.ij} \end{cases},\ \forall\, i\in\mathcal{N}_z,\ ij\in\mathcal{M}_z,\ z\in\mathcal{Z} \tag{1d}$$

$$\boldsymbol{u}_{z.i}^{t} \in \Omega_{z.i},\ \forall\, i\in\mathcal{N}_z^{+},\ z\in\mathcal{Z} \tag{1e}$$

where $t\in\mathcal{T}$ is an arbitrary time instant. (1a) captures the objective to minimize the power losses and operation costs of DERs; f_z^t denotes the operation costs of DERs of each area, and κ is a weight coefficient related to the power losses. (1b)–(1c) are the power flow equations composed of the internal part (1b) [37] and the boundary constraint (1c). (1d) describes the service limits on voltages and currents; (1e) describes the physical constraint of DERs; for example, the PV outputs are limited by weather conditions and the power ratings jointly, and the BESS outputs are limited by the power ratings and the stored energy jointly (see Appendix A for more details about the constraints of DERs).

Four points should be noted. First, for ease of exposition, we model the OPF problem in a special setup where each node (excluding the root node of area 0) is connected with a DER; however, the algorithm proposed next can be easily extended to arbitrary setups without any limits on the locations and numbers of DERs. Second, (1b) is formulated with ignoring shunt capacitance since the lines in distribution systems are typically short. Third, we consider a distribution system that has a radial topology; this accords with the real applications that distribution systems always operate in radial structures although they are designed as meshed networks. Finally, we neglect the dynamics of distribution systems, in which we model the loads as the constant power models and assume that the outputs of DERs are equal to their set-points.

3. Algorithm

This section presents a distributed online algorithm to solve the OPF problem (1). Section 3.1 gives the outline of the proposed algorithm; Sections 3.2 and 3.3 explicate some key details of the algorithm; Section 3.4 gives the procedure to implement the algorithm.

3.1. Outline of the distributed online algorithm

We cope with the complex constraint (1d) using the penalty method. Squaring the voltage and current terms, and transforming the constraint (1d) into the objective, one can get a penalty function:

$$L^t = \sum_{z\in\mathcal{Z}} L_z^t = \sum_{z\in\mathcal{Z}} \left\{ f_z^t(\boldsymbol{u}_z^t) + \left\{ \begin{array}{c} [\boldsymbol{U}_z^t - \bar{\boldsymbol{U}}_z]^{+\mathrm{T}}\mathrm{diag}(\bar{\boldsymbol{\gamma}}_z)[\boldsymbol{U}_z^t - \bar{\boldsymbol{U}}_z]^{+} + \\ [\underline{\boldsymbol{U}} - \boldsymbol{U}_z^t]^{+\mathrm{T}}\mathrm{diag}\left(\underline{\boldsymbol{\gamma}}_z\right)[\underline{\boldsymbol{U}} - \boldsymbol{U}_z^t]^{+} + \\ [\boldsymbol{l}_z^t - \bar{\boldsymbol{l}}_z]^{+\mathrm{T}}\mathrm{diag}(\boldsymbol{\gamma}_z)[\boldsymbol{l}_z^t - \bar{\boldsymbol{l}}_z]^{+} \end{array} \right\} \right\} \tag{2}$$

with reformulating the OPF model (1) as

$$\mathrm{Min}\ L^t \tag{3a}$$

s.t. (1e), and

$$\begin{cases} \boldsymbol{U}_z^t = \boldsymbol{U}_z^t(\boldsymbol{u}_0^t,\ldots,\boldsymbol{u}_z^t,\ldots) \\ \boldsymbol{l}_z^t = \boldsymbol{l}_z^t(\boldsymbol{u}_0^t,\ldots,\boldsymbol{u}_z^t,\ldots) \end{cases},\ z\in\mathcal{Z} \tag{3b}$$

where the constraint (3b) is equivalent to the power flow equations composed of (1b) and (1c).

Thus, we get a time-varying OPF problem that only contains the power flow constraints and the DER output limits. The reformulated problem can be solved by an online algorithm (see the Algorithm 1 in [17]) based on measurements. In each time interval $[t-\Delta t,\ t]$, it solves the Taylor approximation of the OPF model (3):

$$\min_{\boldsymbol{u}_z^t,z\in\mathcal{Z}} \widetilde{L}^t(\ldots,\boldsymbol{u}_{z-1}^t,\boldsymbol{u}_z^t,\boldsymbol{u}_{z+1}^t,\ \ldots) = \frac{1}{2}\sum_{s\in\mathcal{Z}}\sum_{z\in\mathcal{Z}}[(\boldsymbol{u}_s^t - \boldsymbol{u}_s^{t.0})^{\mathrm{T}}\boldsymbol{B}_{sz}^t(\boldsymbol{u}_z^t - \boldsymbol{u}_z^{t.0})] + \sum_{z\in\mathcal{Z}}(\boldsymbol{g}_z^t)^{\mathrm{T}}(\boldsymbol{u}_z^t - \boldsymbol{u}_z^{t.0}) \tag{4}$$

s.t. (1e), (3b)

with applying the solution u_z^{t*} to regulate the DER set-points at each instant t. $u_z^{t,0}$ is the base value for the approximation at t and takes the value of the DER outputs at t − Δt. g_z^t and B_{sz}^t are as

$$\boldsymbol{g}_z^t = \frac{\partial L^t}{\partial \boldsymbol{u}_z^t}, \quad \boldsymbol{B}_{sz}^t = \frac{\partial^2 L^t}{\partial \boldsymbol{u}_s^t \partial \boldsymbol{u}_z^t}. \tag{5}$$

The online algorithm above use measurements to cope with the power flow constraints. Expanding g_z^t and B_{sz}^t, one can find that they are combinations of power flow states and the corresponding sensitivities (the complete expanded results are shown in Section 3.2). Here the power flow states are obtained based on measurements. It utilizes the distribution grid as a natural power flow solver that feeds back the power flow states automatically. This saves significant time compared with some traditional OPF algorithms that obtain the power flow states by solving the non-convex power flow equations.

However, the online algorithm mentioned above is centralized. We furtherly utilize a distributed solution (see the Algorithm 4.1 in [38]) to solve (4); in each interval $[t - \Delta t, t]$, each area ($z \in Z$) iteratively solves an individual problem:

$$\begin{aligned} &\min_{\boldsymbol{u}_z^{t,k}} \widetilde{L}_z^{t,k}(...,\bar{\boldsymbol{u}}_{z-1}^{t,k-1},\boldsymbol{u}_z^{t,k},\bar{\boldsymbol{u}}_{z+1}^{t,k-1}, ...) = \\ &\frac{1}{2}(\boldsymbol{u}_z^{t,k} - \boldsymbol{u}_z^{t,0})^{\mathrm{T}} \boldsymbol{B}_{zz}^t (\boldsymbol{u}_z^{t,k} - \boldsymbol{u}_z^{t,0}) + \\ &\frac{1}{2}\sum_{s\in\mathcal{Z},s\neq z} (\bar{\boldsymbol{u}}_s^{t,k-1} - \boldsymbol{u}_s^{t,0})^{\mathrm{T}} \boldsymbol{B}_{sz}^t (\boldsymbol{u}_z^{t,k} - \boldsymbol{u}_z^{t,0}) + (\boldsymbol{g}_z^t)^{\mathrm{T}} (\boldsymbol{u}_z^{t,k} - \boldsymbol{u}_z^{t,0}) \end{aligned} \tag{6a}$$

s.t. (1e) and

$$\boldsymbol{u}_{z.i}^{t,k} \in \Omega_{\mathrm{step},z.i}^t \tag{6b}$$

with an update:

$$\bar{\boldsymbol{u}}_z^{t,k} = \omega_z \boldsymbol{u}_z^{t,k*} + (1 - \omega_z)\bar{\boldsymbol{u}}_z^{t,k-1} \tag{6c}$$

where $u_z^{t,k*}$ is the solution of (6a)–(6b), and ω_z is a diagonal coefficient matrix whose elements are all in the scope of 0 to 1. (6b) is a step constraint for limiting the set-point change at each time interval, and

$$\Omega_{\mathrm{step},z.i}^t = [P_{z.i}^{t,0} - \Delta P_{z.i}^t, P_{z.i}^{t,0} + \Delta P_{z.i}^t] \times [Q_{z.i}^{t,0} - \Delta Q_{z.i}^t, Q_{z.i}^{t,0} + \Delta Q_{z.i}^t]$$

where $[\Delta P_{z.i}^t, \Delta Q_{z.i}^t]$ is the step length. This constraint can limit the changes of DER outputs to avoid obvious violations of voltages and currents.

Now, it can be seen that each area only decides on the set-points of the internal DERs. Moreover, each iteration based on (6) produces a feasible solution that satisfies the power flow equations, which means that the iterations can stop at any steps. It is significant for the applications on fast time scales, since the online applications may force the iterations to stop without convergence. Using the iterations based on (6), one can always get a feasible solution that can be applied to electrical grids.

The iterative solution based on (6) gives an outline of the distributed online algorithm, but two key points are still required to be addressed. First, the formulation of $\widetilde{L}_z^{t,k}$ still needs a centralized way; it requires to collect measurements and parameters across the whole considered distribution system and to calculate the sensitivities of power flow states in a centralized way. Second, the DER outputs are jointly limited by the physical constraint (1e) and the step constraint (6b); the joint constraints are so complex that (6) is unable to be solved fast enough to satisfy the requirement of the applications on fast time scales. Thus, we propose a method to formulate $\widetilde{L}_z^{t,k}$ in a distributed way and a strategy to simplify the coupling constraints on DER outputs next.

3.2. Distributed calculation of $\widetilde{L}_z^{t,k}$

The key to formulate $\widetilde{L}_z^{t,k}$ for each area ($z \in \mathcal{Z}$) is to calculate $\boldsymbol{g}_z^t$, $\boldsymbol{B}_{zz}^t$ and $\sum_{s\in\mathcal{Z},s\neq z} (\boldsymbol{u}_s^{t,k-1} - \boldsymbol{u}_s^{t,0})^{\mathrm{T}} \boldsymbol{B}_{sz}^t$. Set $\boldsymbol{\varphi}_z^{t,k} = \sum_{s\in\mathcal{Z},s\neq z} (\boldsymbol{u}_s^{t,k-1} - \boldsymbol{u}_s^{t,0})^{\mathrm{T}} \boldsymbol{B}_{sz}^t$ for exposition simplicity.

We first give the principle to decouple the calculation of $\boldsymbol{g}_z^t$, $\boldsymbol{B}_{zz}^t$ and $\boldsymbol{B}_{sz}^t$ ($s \in \mathcal{Z}, s \neq z$) among different areas.

a. For the upper area ($z = 0$), $\boldsymbol{g}_0^t$, $\boldsymbol{B}_{00}^t$ and $\boldsymbol{B}_{s0}^t$ ($s \in \mathcal{Z}^+$) can be expanded as

$$\boldsymbol{g}_0^t = \frac{\partial L_0^t}{\partial \boldsymbol{u}_0} + \sum_{s\in\mathcal{Z}^+} \frac{\partial L_s^t}{\partial \boldsymbol{u}_0} \tag{7a}$$

$$\boldsymbol{B}_{00}^t = \frac{\partial^2 L_0^t}{\partial \boldsymbol{u}_0^2} + \sum_{s\in\mathcal{Z}^+} \frac{\partial^2 L_s^t}{\partial \boldsymbol{u}_0^2} \tag{7b}$$

$$\boldsymbol{B}_{s0}^t = \frac{\partial^2 L_0^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} + \frac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} + \sum_{p\in\mathcal{Z}^+,p\neq s} \frac{\partial^2 L_p^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} \tag{7c}$$

where $\partial L_0^t/\partial u_0$ and $\partial^2 L_0^t/\partial u_0^2$ can be calculated based on the internal information of area 0 (see Appendix C for the calculation). The other terms in the right-hand side of (7) can be decoupled as (see Appendix B for the deduction)

$$\begin{cases} \dfrac{\partial L_s^t}{\partial \boldsymbol{u}_0} = \dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_0} \dfrac{\partial L_s^t}{\partial U_{s.0}} \\ \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_0^2} = \dfrac{\partial^2 L_s^t}{\partial U_{s.0}^2} \dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_0} \left[\dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_0}\right]^{\mathrm{T}} \end{cases}, s \in \mathcal{Z}^+ \tag{8a}$$

$$\begin{cases} \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} = \dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^t}{\partial \boldsymbol{u}_s} \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_{s.\mathrm{in}} \partial \boldsymbol{u}_0} \\ \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} = \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial U_{s.0}} \left[\dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_0}\right]^{\mathrm{T}} \\ \dfrac{\partial^2 L_p^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_0} = \dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^t}{\partial \boldsymbol{u}_s} \dfrac{\partial U_{p.0}^t}{\partial \boldsymbol{u}_{s.\mathrm{in}}} \left[\dfrac{\partial U_{p.0}^t}{\partial \boldsymbol{u}_0}\right]^{\mathrm{T}} \dfrac{\partial^2 L_p^t}{\partial U_{p.0}^2} \end{cases} \begin{matrix} s \in \mathcal{Z}^+ \\ , p \in \mathcal{Z}^+ \\ s \neq p \end{matrix} \tag{8b}$$

b. For each lower area ($z \in \mathcal{Z}^+$), $\boldsymbol{g}_z^t$ and $\boldsymbol{B}_{zz}^t$, $\boldsymbol{B}_{sz}^t$ ($s \in \mathcal{Z}, s \neq z$) can be expanded as

$$\boldsymbol{g}_z^t = \frac{\partial L_z^t}{\partial \boldsymbol{u}_z} + \frac{\partial L_0^t}{\partial \boldsymbol{u}_z} + \sum_{s\in\mathcal{Z}^+,s\neq z} \frac{\partial L_s^t}{\partial \boldsymbol{u}_z} \tag{9a}$$

$$\boldsymbol{B}_{zz}^t = \frac{\partial^2 L_z^t}{\partial \boldsymbol{u}_z^2} + \frac{\partial^2 L_0^t}{\partial \boldsymbol{u}_z^2} + \sum_{s\in\mathcal{Z}^+,s\neq z} \frac{\partial^2 L_s^t}{\partial \boldsymbol{u}_z^2} \tag{9b}$$

$$\boldsymbol{B}_{sz}^t = \begin{cases} \textit{the transposition of } \boldsymbol{B}_{z0}^t, & s = 0 \\ \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} + \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} + \dfrac{\partial^2 L_z^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} + \sum_{p\in\mathcal{Z}^+,p\neq s,p\neq z} \dfrac{\partial^2 L_p^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z}, & \begin{matrix} s \in \mathcal{Z}^+ \\ s \neq z \end{matrix} \end{cases} \tag{9c}$$

where $\partial L_z^t/\partial \boldsymbol{u}_z$ and $\partial^2 L_z^t/\partial u_z^2$ can be calculated based on the internal information of each lower area (see Appendix C for the calculation). The other terms in the right-hand side of (9) can be decoupled as (see Appendix B for the deduction)

$$\begin{cases} \dfrac{\partial L_0^t}{\partial \boldsymbol{u}_z} = \dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z} \dfrac{\partial L_0^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}} \\ \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_z^2} = \dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z} \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}^2} \left[\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z}\right]^{\mathrm{T}} \\ \dfrac{\partial L_s^t}{\partial \boldsymbol{u}_z} = \dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z} \dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}} \dfrac{\partial L_s^t}{\partial U_{s.0}} \\ \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_z^2} = \dfrac{\partial^2 L_s^t}{\partial U_{s.0}^2} \dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z} \dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}} \left[\dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}}\right]^{\mathrm{T}} \left[\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z}\right]^{\mathrm{T}} \end{cases}, \begin{matrix} s \in \mathcal{Z}^+ \\ s \neq z \end{matrix} \tag{10a}$$

$$\begin{cases} \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} = \dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^t}{\partial \boldsymbol{u}_s} \dfrac{\partial^2 L_0^t}{\partial \boldsymbol{u}_{s.\mathrm{in}} \partial \boldsymbol{u}_{z.\mathrm{in}}} \left[\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z}\right]^{\mathrm{T}} \\ \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} = \dfrac{\partial^2 L_s^t}{\partial \boldsymbol{u}_s \partial U_{s.0}} \left[\dfrac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}}\right]^{\mathrm{T}} \left[\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z}\right]^{\mathrm{T}} \\ \dfrac{\partial^2 L_z^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} = \dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^t}{\partial \boldsymbol{u}_s} \dfrac{\partial U_{z.0}^t}{\partial \boldsymbol{u}_{s.\mathrm{in}}} \left[\dfrac{\partial^2 L_z^t}{\partial \boldsymbol{u}_z \partial U_{z.0}}\right]^{\mathrm{T}} \\ \dfrac{\partial^2 L_p^t}{\partial \boldsymbol{u}_s \partial \boldsymbol{u}_z} = \dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^t}{\partial \boldsymbol{u}_s} \dfrac{\partial U_{p.0}^t}{\partial \boldsymbol{u}_{s.\mathrm{in}}} \left[\dfrac{\partial U_{p.0}^t}{\partial \boldsymbol{u}_{z.\mathrm{in}}}\right]^{\mathrm{T}} \left[\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^t}{\partial \boldsymbol{u}_z}\right]^{\mathrm{T}} \dfrac{\partial^2 L_p^t}{\partial U_{p.0}^2} \end{cases} \begin{matrix} s \in \mathcal{Z}^+ \\ p \in \mathcal{Z}^+ \\ s \neq p \\ s \neq z \end{matrix} \tag{10b}$$

Then, by multiplying $\boldsymbol{B}_{sz}^{t}$ by $\boldsymbol{u}_{s}^{t.k-1}-\boldsymbol{u}_{s}^{t.0}$ $(s\in\mathcal{Z}, s\neq z)$, one can obtain $\boldsymbol{\varphi}_{z}^{t.k}$.

The decoupled results (7)–(10) lay a foundation for calculating $\boldsymbol{B}_{zz}^{t}$, $\boldsymbol{g}_{z}^{t}$ and $\boldsymbol{\varphi}_{z}^{t.k}$ in a distributed way, since all the terms in the right-hand side of (7)–(10) can be calculated only based on the internal information of each individual area(see Appendix C for the calculation method). After calculating the corresponding terms, all the areas require to exchange and recombine these terms furtherly to formulate $\boldsymbol{B}_{zz}^{t}$, $\boldsymbol{g}_{z}^{t}$ and $\boldsymbol{\varphi}_{z}^{t.k}$. To sum up, we give a strategy to calculate $\boldsymbol{B}_{zz}^{t}$, $\boldsymbol{g}_{z}^{t}$ and $\boldsymbol{\varphi}_{z}^{t.k}$ as follows.

Strategy 1: distributed calculation of $\boldsymbol{B}_{zz}^{t}$, $\boldsymbol{g}_{z}^{t}$ and $\boldsymbol{\varphi}_{z}^{t.k}$

a. For each lower area ($z\in\mathcal{Z}^{+}$), it calculates (see Appendix C for the calculation)

$$\frac{\partial L_{z}^{t}}{\partial \boldsymbol{u}_{z}},\ \frac{\partial^{2} L_{z}^{t}}{\partial \boldsymbol{u}_{z}^{2}},\ \frac{\partial \boldsymbol{u}_{z.\mathrm{in}}^{t}}{\partial \boldsymbol{u}_{z}},\ \frac{\partial L_{z}^{t}}{\partial U_{z.0}},\ \frac{\partial^{2} L_{z}^{t}}{\partial U_{z.0}^{2}},\ \frac{\partial^{2} L_{z}^{t}}{\partial \boldsymbol{u}_{z}\partial U_{z.0}} \tag{11}$$

and then it multiplies $\boldsymbol{u}_{z}^{t.k-1}-\boldsymbol{u}_{z}^{t.0}$ by $\partial \boldsymbol{u}_{z.\mathrm{in}}^{t}/\partial \boldsymbol{u}_{z}$ and $\partial^{2} L_{z}^{t}/\partial \boldsymbol{u}_{z}\partial U_{z.0}$ respectively to formulate the combined terms:

$$\begin{cases}[\boldsymbol{u}_{z}^{t.k-1}-\boldsymbol{u}_{z}^{t.0}]^{\mathrm{T}}\dfrac{\partial \boldsymbol{u}_{z.\mathrm{in}}^{t}}{\partial \boldsymbol{u}_{z}}\\ [\boldsymbol{u}_{z}^{t.k-1}-\boldsymbol{u}_{z}^{t.0}]^{\mathrm{T}}\dfrac{\partial^{2} L_{z}^{t}}{\partial \boldsymbol{u}_{z}\partial U_{z.0}}\end{cases} \tag{12}$$

and sends $\partial L_{z}^{t}/\partial U_{z.0}$, $\partial^{2} L_{z}^{t}/\partial U_{z.0}^{2}$ and (12) to the upper area.

b. For the upper area (z = 0), it calculates (see Appendix C for the calculation)

$$\frac{\partial L_{0}^{t}}{\partial \boldsymbol{u}_{0}},\ \frac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{0}^{2}},\ \frac{\partial U_{s.0}^{t}}{\partial \boldsymbol{u}_{0}},\ \frac{\partial U_{s.0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}},\ \frac{\partial L_{0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}},\ \frac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}^{2}},\ \frac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}\partial \boldsymbol{u}_{0}},\ \frac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}\partial \boldsymbol{u}_{s.\mathrm{in}}}$$

$$p\in\mathcal{Z}^{+}, s\in\mathcal{Z}^{+}, p\neq s \tag{13}$$

and combines the received terms and (13) to formulate $\boldsymbol{g}_{0}^{t}$, $\boldsymbol{B}_{00}^{t}$, $\boldsymbol{\varphi}_{0}^{t.k}$ based on (7) and (8). As well, it formulates the following combined terms:

$$\begin{cases}\sum_{p\in\mathcal{Z}^{+},p\neq s}\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\dfrac{\partial L_{p}^{t}}{\partial U_{p.0}}\\ \sum_{p\in\mathcal{Z}^{+},p\neq s}\dfrac{\partial^{2} L_{p}^{t}}{\partial U_{p.0}^{2}}\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\left[\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\right]^{\mathrm{T}}\end{cases},\ s\in\mathcal{Z}^{+} \tag{14a}$$

$$\begin{cases}[\boldsymbol{u}_{0}^{t.k-1}-\boldsymbol{u}_{0}^{t.0}]^{\mathrm{T}}\dfrac{\partial U_{s.0}^{t}}{\partial \boldsymbol{u}_{0}}\\ [\boldsymbol{u}_{0}^{t.k-1}-\boldsymbol{u}_{0}^{t.0}]^{\mathrm{T}}\dfrac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{0}\partial \boldsymbol{u}_{s.\mathrm{in}}}\\ [\boldsymbol{u}_{0}^{t.k-1}-\boldsymbol{u}_{0}^{t.0}]^{\mathrm{T}}\sum_{p\in\mathcal{Z}^{+},p\neq s,}\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{0}}\left[\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\right]^{\mathrm{T}}\dfrac{\partial^{2} L_{p}^{t}}{\partial U_{p.0}^{2}}\end{cases},\ s\in\mathcal{Z}^{+} \tag{14b}$$

$$\begin{cases}[\boldsymbol{u}_{s}^{t.k-1}-\boldsymbol{u}_{s}^{t.0}]^{\mathrm{T}}\dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^{t}}{\partial \boldsymbol{u}_{s}}\dfrac{\partial^{2} L_{0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}\partial \boldsymbol{u}_{p.\mathrm{in}}}\\ [\boldsymbol{u}_{s}^{t.k-1}-\boldsymbol{u}_{s}^{t.0}]^{\mathrm{T}}\dfrac{\partial^{2} L_{s}^{t}}{\partial \boldsymbol{u}_{s}\partial V_{s.0}}\left[\dfrac{\partial U_{s.0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}}\right]^{\mathrm{T}}\\ [\boldsymbol{u}_{s}^{t.k-1}-\boldsymbol{u}_{s}^{t.0}]^{\mathrm{T}}\dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^{t}}{\partial \boldsymbol{u}_{s}}\dfrac{\partial U_{p.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\\ [\boldsymbol{u}_{s}^{t.k-1}-\boldsymbol{u}_{s}^{t.0}]^{\mathrm{T}}\dfrac{\partial \boldsymbol{u}_{s.\mathrm{in}}^{t}}{\partial \boldsymbol{u}_{s}}\sum_{v\in\mathcal{Z}^{+},v\neq s,v\neq z}\dfrac{\partial U_{v.0}^{t}}{\partial \boldsymbol{u}_{s.\mathrm{in}}}\left[\dfrac{\partial U_{v.0}^{t}}{\partial \boldsymbol{u}_{p.\mathrm{in}}}\right]^{\mathrm{T}}\dfrac{\partial^{2} L_{v}^{t}}{\partial U_{v.0}^{2}}\end{cases}\begin{matrix}\\ \\ p\in\mathcal{Z}^{+}\\ ,\ s\in\mathcal{Z}^{+}\\ s\neq p\end{matrix} \tag{14c}$$

and sends $\partial L_{0}^{t}/\partial \boldsymbol{u}_{s.\mathrm{in}}$, $\partial^{2} L_{0}^{t}/\partial \boldsymbol{u}_{s.\mathrm{in}}^{2}$ and to lower areas.

c. Each lower area combines the received terms and (11) to formulate $\boldsymbol{B}_{zz}^{t}$, $\boldsymbol{g}_{z}^{t}$ and $\boldsymbol{\varphi}_{z}^{t.k}$ based on (9) and (10).

Note that in the designed strategy, $[\boldsymbol{u}_{z}^{t.k-1}-\boldsymbol{u}_{z}^{t.0}]^{\mathrm{T}}(z\in\mathcal{Z})$ and the related sensitivities are appropriately combined as a combination in the process of information transmission. This can reduce the amount of information transmitted among different areas.

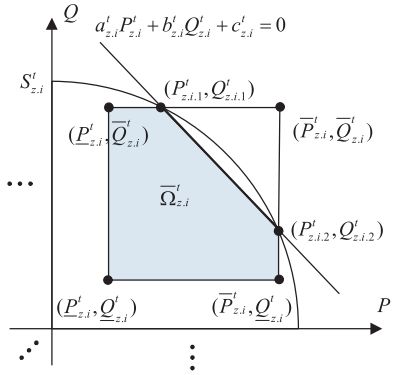


Fig. 2. Sketch of the polygon constraint on DER outputs.

3.3. Simplification of the constraints on DER outputs

The objective is to simplify the coupling constraints composed of (1e) and (6b). (1e) can be split into two parts: the first part is the limit on DER active outputs (e.g., maximum active outputs of PVs limited by weather conditions); the second part is the inverter circle constraint (See Appendix A for the details). By adding the active output limit, the first part of (1e), to the constraint (6b), one can obtain a unified box constraint for each DER. Then, adjusting the unified box constraint adaptively to the inverter circle constraint, the second part of (1e), one can get a polygon constraint for each DER shown as Fig. 2 and (15).

$$\boldsymbol{u}_{z.i}^{t}\in\bar{\Omega}_{z.i}^{t}=\left\{\begin{array}{c}\underline{P}_{z.i}^{t}\leqslant P_{z.i}^{t}\leqslant\bar{P}_{z.i}^{t}\\ \underline{Q}_{z.i}^{t}\leqslant Q_{z.i}^{t}\leqslant\bar{Q}_{z.i}^{t}\\ a_{z.i}^{t}P_{z.i}^{t}+b_{z.i}^{t}Q_{z.i}^{t}+c_{z.i}^{t}\leqslant 0\end{array}\right\} \tag{15}$$

where $\bar{P}_{z.i}^{t}$, $\bar{Q}_{z.i}^{t}$, $\underline{P}_{z.i}^{t}$, $\underline{Q}_{z.i}^{t}$ are

$$\begin{cases}\bar{P}_{z.i}^{t}=\min\{P_{z.i}^{t.0}+\Delta P_{z.i}^{t},\ \bar{P}_{z.i}^{\mathrm{ph}.t}\},\ \bar{Q}_{z.i}^{t}=Q_{z.i}^{t.0}+\Delta Q_{z.i}^{t}\\ \underline{P}_{z.i}^{t}=\max\{P_{z.i}^{t.0}-\Delta P_{z.i}^{t},\ \underline{P}_{z.i}^{\mathrm{ph}.t}\},\ \underline{Q}_{z.i}^{t}=Q_{z.i}^{t.0}-\Delta Q_{z.i}^{t}\end{cases} \tag{16}$$

and $a_{z.i}^{t}$, $b_{z.i}^{t}$, $c_{z.i}^{t}$ are coefficients determined by the relative position between the unified box constraint and the inverter circle constraint. One can classify this relative position into several types and then calculate the coefficients based on geometry methods. For exposition simplicity, we only illustrate one case of the position types and give the corresponding calculation for $a_{z.i}^{t}$, $b_{z.i}^{t}$, $c_{z.i}^{t}$ visualized as Fig. 2 and (17). For the other position types, one can also deduce similar results.

$$\begin{cases}a_{z.i}^{t}=Q_{z.i.1}^{t}-Q_{z.i.2}^{t},\ b_{z.i}^{t}=P_{z.i.2}^{t}-P_{z.i.1}^{t}\\ c_{z.i}^{t}=P_{z.i.1}^{t}(Q_{z.i.2}^{t}-Q_{z.i.1}^{t})-Q_{z.i.1}^{t}(P_{z.i.2}^{t}-P_{z.i.1}^{t})\end{cases} \tag{17}$$

where

$$\begin{cases}P_{z.i.1}^{t}=\sqrt{(S_{z.i}^{t})^{2}-(\bar{Q}_{z.i}^{t})^{2}},\ Q_{z.i.1}^{t}=\bar{Q}_{z.i}^{t}\\ P_{z.i.2}^{t}=\bar{P}_{z.i}^{t},\ Q_{z.i.2}^{t}=\sqrt{(S_{z.i}^{t})^{2}-(\bar{P}_{z.i}^{t})^{2}}\end{cases} \tag{18}$$

In formulating the simplified constraints, $\Delta P_{z.i}^{t}$, $\Delta Q_{z.i}^{t}$ are adjusted based on a strategy represented in Appendix D. The strategy aims to decrease the step length when the voltages or currents are adjacent to the boundaries, or when the objective is adjacent to the optimal value. This can restrain the oscillation of the voltages and currents around the boundaries and the oscillation of the objective around the optimal value.

3.4. Procedure to implement the algorithm

We give the procedure to implement the distributed online OPF algorithm in Algorithm 1. An outer loop iterates and applies the iterative result to regulate the DER set-points, where the iterative result

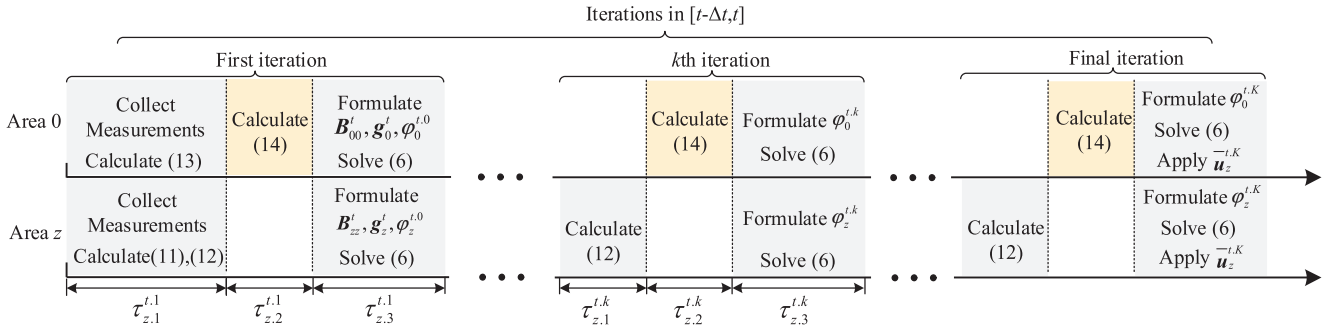


Fig. 3. Sequence diagram for the implementation in each time interval.

is obtained by an inner loop that iterates K times in the interval $[t - \Delta t, t]$. With the two-layer loops, the DER outputs can be driven to a neighborhood of the solutions of the time-varying OPF problem (3) gradually, which will be investigated by the case studies.

Algorithm 1: *distributed online OPF algorithm*

For each area in each interval $[t - \Delta t, t]$,
(1) Collect voltage and current measurements.
(2) Formulate $\boldsymbol{B}_{zz}^t$, $\boldsymbol{g}_z^t$ based on the strategy 1.
(3) For $k = 1$ to K
a. Formulate $\varphi_z^{t,k}$ based on the strategy 1.
b. Solve (6) to obtain $\bar{\boldsymbol{u}}_z^{t,k}$.
c. Return to Step a.
(4) Apply $\bar{\boldsymbol{u}}_z^{t,K}$ to adjust the DER set-points.
(5) Wait until the next time interval and go to Step 1.

The sequence diagram for the implementation in each interval is as Fig. 3. The calculation in each interval contains several iterations, and each iteration contains three fragments. Only the first fragment of the first iteration handles the collection of measurements and the calculation of (11) and (13), and only the final fragment of the final iteration handles the application of the iterative result to adjust the DER set-points. Based on the sequence diagram, we deduce the minimum calculation time for each update of the DER set-points as

$$\tau^t = \sum_{k=1}^{K} \{\mathrm{Max}\{\tau_{z.1}^{t,k}\}_{z\in\mathcal{Z}} + \tau_{0.2}^{t,k} + \mathrm{Max}\{\tau_{z.3}^{t,k}\}_{z\in\mathcal{Z}}\} \tag{19}$$

4. Case studies

The validity of the algorithm is verified through numerical simulation analysis in four aspects: (1) the convergence of the proposed algorithm, (2) the satisfaction of the constraints on voltages and currents, (3) the satisfaction of the operational constraints of DERs, (4) the calculation time. We program the algorithm using MATLAB 9.1 on a computer with an Intel Xeon (E3-1220 V2, 3.10 GHz) CPU and 8 GB RAM. The measurements are produced using MATPOWER 6.0, and the individual approximate problem (6) is solved by GUROBI 8.1.1.

4.1. Test systems and settings

Consider a distribution system composed of four IEEE 33-bus systems shown in Fig. 4, where the three lower areas are connected to nodes 13, 22, 25 of the upper area respectively. Each load-bearing node integrates a 100 kW PV system and a 5 kW/15kWh BESS. The base values are 1Mvar and 12.66 kV. Time-varying loads and PV active outputs are considered as shown in Figs. 5 and 6. The power profiles are generated based on the datasets of OPENDSS. The raw data of loads and PV outputs are on a 60-second basis and a two-second basis respectively. We add variations to the raw data of loads to unify the time granularity to two seconds. f_z^t is set as

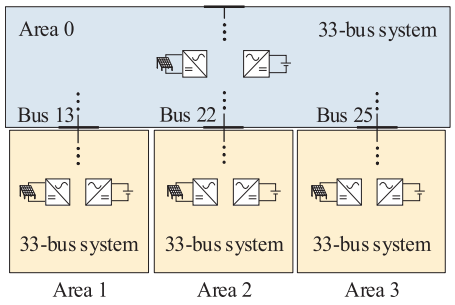


Fig. 4. Distribution system composed of four IEEE 33-bus systems.

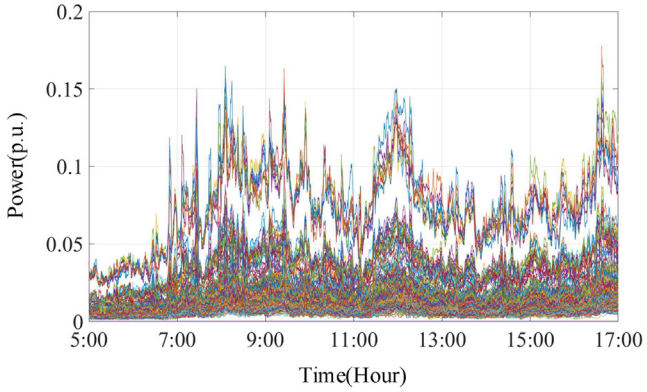


Fig. 5. Profiles of loads.

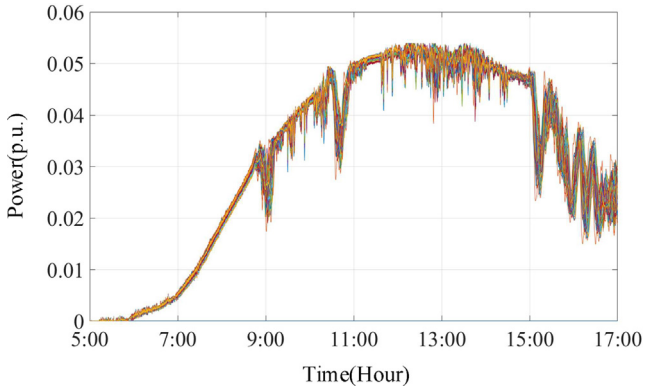
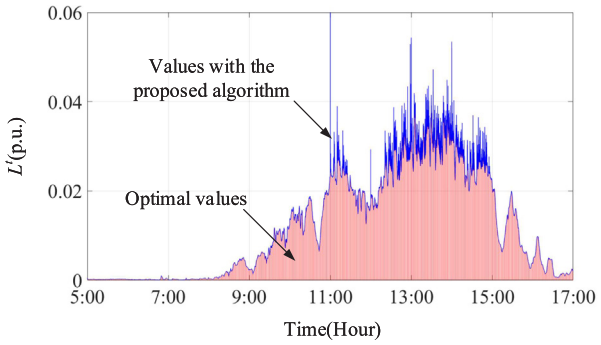


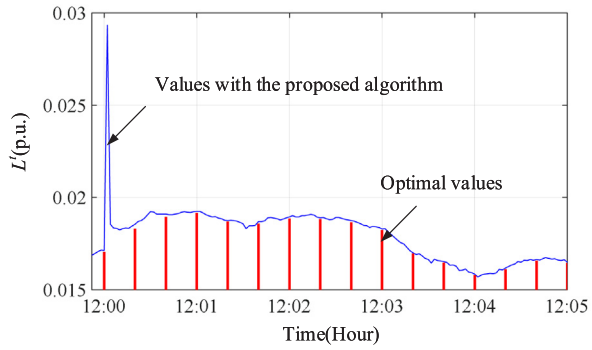
Fig. 6. Profiles of active outputs of PVs.

$$f_z^t = \sum_{i\in\mathcal{N}_z^t} C_{z.i}^t(\boldsymbol{u}_{z.i}^t)$$

where $C_{z.i}^t(\boldsymbol{u}_{z.i}^t)$ denotes the operation costs of DERs. For PVs,



(a) Complete profile of L^t changes



(b) A portion of the profile of L^t changes

Fig. 7. OPF tracking results.

$$C^t_{z.i}(\boldsymbol{u}^t_{z.i}) = a^{\text{pv}}_{z.i}(P^t_{z.i} - \bar{P}^{ph.t}_{z.i})^2 + b^{\text{pv}}_{z.i}(Q^t_{z.i})^2,$$

and for BESSs,

$$C^t_{z.i}(\boldsymbol{u}^t_{z.i}) = a^{\text{b}}_{z.i}(P^t_{z.i})^2 + b^{\text{b}}_{z.i}(Q^t_{z.i})^2,$$

The related parameters are set in per unit as follows. $a^{\text{pv}}_{z.i} = 1$, $b^{\text{pv}}_{z.i} = 0.01$, $a^{\text{b}}_{z.i} = 0.25$, $Q^t_{z.i} = 0.01$, and $\bar{\gamma}_{z.i} = \underline{\gamma}_{z.i} = 10$, $\gamma_{z.ij} = 0.1$, and $\kappa = 0.1$, and $\varepsilon_{\text{U}} = 0.001$, $\zeta_{\text{U}} = 0.01$, $\varepsilon_l = 0.5$, $\varepsilon_{\text{loss}} = 10$–$4$ (in Appendix D). $\Delta P_{\max}$, $\Delta Q_{\max}$ and $\Delta P_{\min}$, $\Delta Q_{\min}$ (in Appendix D) are set as 0.1 and 0.005 times of the rated powers of the related inverters respectively. The upper and lower limits on voltage are set to 1.05 and 0.95 respectively. For the initial values: for each PV system, the active output is set based on the principle of maximum power point tracking, and the reactive output is 0; for each BESS, the active and reactive outputs are both 0. The DER set-points are updated every two seconds ($\Delta t = 2$ s), in which one iteration is performed ($K = 1$).

4.2. Test results

(1) Convergence of the proposed algorithm

Fig. 7 shows the OPF tracking results. The bars are the optimal values of L^t obtained every 20 s. The curve represents the values of L^t achieved by the proposed algorithm. One can see that the achieved objective values sometimes deviate from the optimal values but then converge towards the optimal values soon. The momentary deviations are mainly due to that this paper simulates a cloudy case where the PV active outputs change abruptly. The average gap between the values of L^t achieved by the proposed algorithm and the optimal values is 2.99%.

(2) Satisfaction of the constraints on voltages and currents

Figs. 8 and 9, Figs. 10 and 11 show the profiles of voltages and currents respectively. Without any regulation, the voltage and current magnitudes exceed the upper limit seriously from 10:00 to 15:00 due to

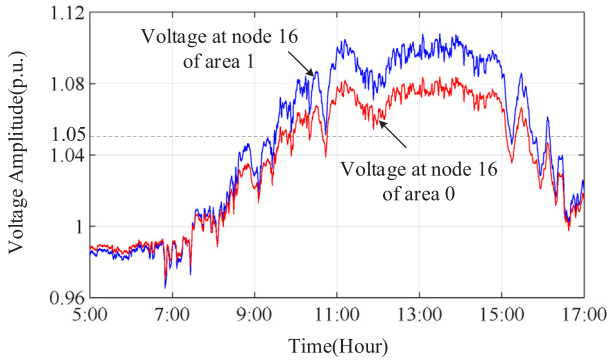
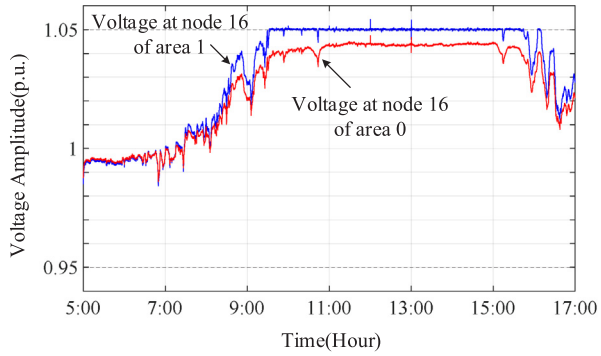
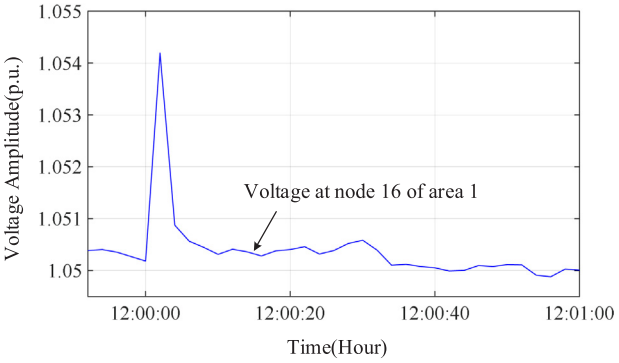


Fig. 8. Voltage profiles at nodes 16 of area 0 and area 1 respectively without regulation.



(a) Complete profiles of the voltage changes



(b) A portion of the profile of the voltage changes

Fig. 9. Voltage profiles at nodes 16 of area 0 and area 1 respectively regulated by the proposed algorithm.

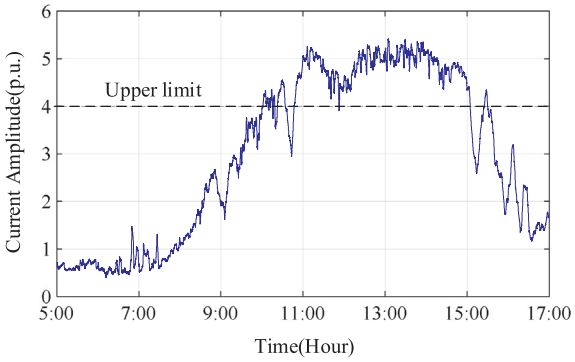


Fig. 10. Current profiles on line (0,1) of area 0 without regulation.

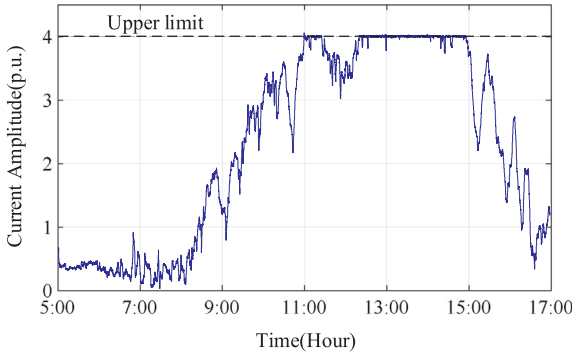


Fig. 11. Current profiles on line (0,1) of area 0 regulated by the proposed algorithm.

the high PV active outputs. In contrast, by utilizing the proposed algorithm, the voltage and current magnitudes are regulated to flatten profiles near the upper limits. In the control process, the voltages experience a few flickers but soon are enforced towards the limited range in a few seconds. A tiny violation of voltages and currents still exists at the flatten part due to the penalty method implemented for coping with the constraints. The largest and average voltage violations of all nodes are 1.0544 p.u. and 1.0502 p.u. respectively, and the largest and average current violations of all lines are 4.054 p.u. and 4.005 p.u. respectively.

(3) Satisfaction of the operational constraints of DERs

Figs. 12 and 13 show the profiles of the outputs and the stored energy of BESSs. One can see that the BESSs are regulated in real time to respond to the operational changes while satisfying the operational constraints on their outputs and energy throughout the control process. In the periods when the PV active outputs are low, the outputs of the BESSs are positive; it is due to that in this time the branch flows are from the upstream to the downstream, and the positive outputs of the BESSs are useful to decrease the power losses. In the periods when the PV active outputs are high, the active outputs of the BESSs change from positive to negative; this can diminish the violations of voltages and currents due to the high active outputs of PVs. Moreover, in the process of charging, the BESSs reach the upper limits on the energy so they stop charging and then the values of the reactive outputs begin to increase to diminish the violations of the voltage and current constraints.

(4) Analysis of the calculation time

Fig. 14 shows the probability density of τ^t, of which the average value is 0.15 s and the maximum is 0.40 s. It shows that the proposed algorithm is computationally efficient since the calculation time is

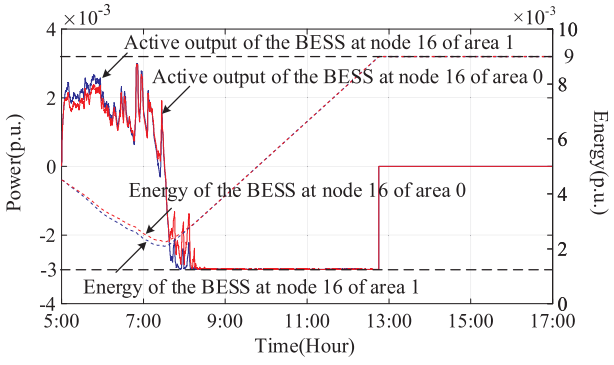


Fig. 12. Active outputs of the BESSs at nodes 16 of area 0 and area 1 respectively.

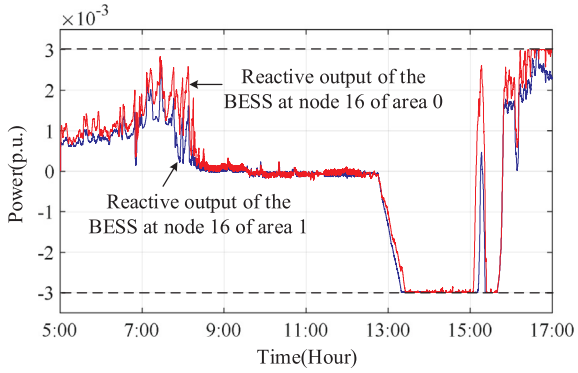


Fig. 13. Reactive outputs of the BESSs at nodes 16 of area 0 and area 1 respectively.

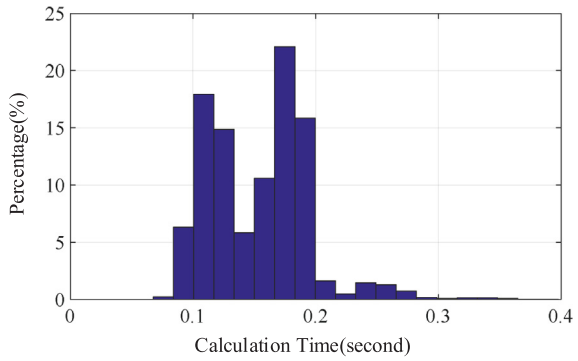


Fig. 14. Probability density of the calculation time for each update of the DER set-points.

consistent with the time interval of two seconds to update the DER set-points.

5. Conclusions

This paper proposed a distributed online OPF algorithm for distribution systems. The algorithm can coordinate the update of DER set-points on fast time scales only depending on local measurements and boundary information for each distribution area. Comparing with the existing distributed online algorithms that require some special setups (e.g., without considering current constraints, or without considering the hard constraints on voltages, or only considering DC power flow models), the proposed algorithm can cope with various setups for the objective functions and constraints. The validity of the algorithm was verified by case studies, which showed that it can continuously drive the operation of DERs towards the time-varying OPF targets composed of power losses and operation costs of DERs, as well as maintain the voltages and currents within predefined limits. It is noted that this work ignored the dynamics of DERs and measurement errors that exist in real physical systems. These matters will be considered in our future works.

CRediT authorship contribution statement

Xingxu Zhu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft. **Xueshan Han:** Supervision, Project administration, Funding acquisition, Writing - review & editing. **Ming Yang:** Writing - review & editing, Visualization. **Yijing Xu:** Writing - review & editing, Visualization. **Shibo Wang:** Writing - review & editing, Visualization.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

Acknowledgments

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Appendix A. Physical constraints of DERs

For PVs, the outputs are limited by weather conditions and the inverter capacities jointly:

$$\boldsymbol{u}_{z.i}^{t} \in \Omega_{z.i}^{t} = \left\{ \begin{array}{c} (P_{z.i}^{t},\, Q_{z.i}^{t})\colon 0 \leqslant P_{z.i}^{t} \leqslant \bar{P}_{z.i}^{ph.t}, \\ (Q_{z.i}^{t})^2 \leqslant (S_{z.i})^2 - (P_{z.i}^{t})^2 \end{array} \right\} \tag{20}$$

where $S_{z.i}$ is the inverter capacity, and $\bar{P}_{z.i}^{ph.t}$ is the upper limit on the active output determined by weather conditions.

For BESSs, the outputs are limited by the power ratings and the stored energy and the inverter capacities jointly:

$$\boldsymbol{u}_{z.i}^{t} \in \Omega_{z.i}^{t} = \left\{ \begin{array}{c} (P_{z.i}^{t},\, Q_{z.i}^{t})\colon \underline{P}_{z.i}^{\mathrm{ph}.t} \leqslant P_{z.i}^{t} \leqslant \bar{P}_{z.i}^{\mathrm{ph}.t} \\ (Q_{z.i}^{t})^2 \leqslant (S_{z.i})^2 - (P_{z.i}^{t})^2 \end{array} \right\} \tag{21}$$

where $\bar{P}_{z.i}^{\mathrm{ph}.t}$, $\underline{P}_{z.i}^{\mathrm{ph}.t}$ are the upper and lower limits on the active outputs respectively:

$$\begin{cases} \bar{P}_{z.i}^{\mathrm{ph}.t} = \min\{P_{z.i}^{\max},\, (E_{z.i}^{t} - E_{z.i}^{\min})\eta_{\mathrm{d}.z.i}/\triangle t\} \\ \underline{P}_{z.i}^{\mathrm{ph}.t} = -\min\{P_{z.i}^{\max},\, (E_{z.i}^{\max} - E_{z.i}^{t})/(\eta_{\mathrm{c}.z.i}\triangle t)\} \end{cases} \tag{22}$$

where $P_{z.i}^{\max}$ is the rating of active power. $E_{z.i}^{\max}$, $E_{z.i}^{\min}$ are the upper and lower limits on the stored energy respectively. $\eta_{\mathrm{c}.z.i}$, $\eta_{\mathrm{d}.z.i}$ are the charging and discharging efficiencies respectively. $E_{z.i}^{t}$ is the stored energy:

$$\begin{cases} E_{z.i}^{t} = E_{z.i}^{t-1} - \eta_{\mathrm{c}.z.i}P_{z.i}^{t},\ P_{z.i}^{t} \leqslant 0 \\ E_{z.i}^{t} = E_{z.i}^{t-1} - P_{z.i}^{t}/\eta_{\mathrm{d}.z.i},\ P_{z.i}^{t} > 0 \end{cases} \tag{23}$$

Similarly, the physical constraints of other types of DERs such as small gas turbines can also be modeled according to their physics. One can see [39,40] for details about this.

Appendix B. *Proof of* (8) and (10)

The proof of (8) and (10) is based on a linearization of the power flow Eq. (1b) (see [37] for the details):

$$\begin{cases} U_{z.j}^{t} = U_{z.i}^{t} - 2(P_{z.ij}^{t}r_{z.ij} + Q_{z.ij}^{t}x_{z.ij}) + l_{z.ij}^{t}(z_{z.ij})^2 \quad (a) \\ P_{z.ij}^{t} = \sum_{h\in\mathcal{H}_{z.j}} P_{z.jh}^{t} + P_{\mathrm{L}.z.j}^{t} - P_{z.j}^{t} + l_{z.ij}^{t}r_{z.ij} \qquad (b) \\ Q_{z.ij}^{t} = \sum_{h\in\mathcal{H}_{z.j}} Q_{z.jh}^{t} + Q_{\mathrm{L}.z.j}^{t} - Q_{z.j}^{t} + l_{z.ij}^{t}x_{z.ij} \qquad (c). \\ l_{z.ij}^{t} = 2(\widetilde{P}_{z.ij}^{t}P_{z.ij}^{t} + \widetilde{Q}_{z.ij}^{t}Q_{z.ij}^{t})/\widetilde{U}_{z.i}^{t} - \widetilde{l}_{z.ij}^{t} \qquad (d) \\ \qquad \forall\, i \in \mathcal{N}_z,\ ij \in \mathcal{M}_z,\ z \in \mathcal{Z} \end{cases} \tag{24}$$

We take the decomposition of $\partial L_s^t/\partial \boldsymbol{u}_z (s \in \mathcal{Z}^+,\ z \in \mathcal{Z}^+,\ z \neq s)$ as an example. One has

$$P_{\mathrm{loss}}^{t} = \sum_{z\in\mathcal{Z}} \boldsymbol{l}_z^{t\mathrm{T}} r_z \tag{25}$$

and thus $\partial L_s^t/\partial \boldsymbol{u}_z$ can be expanded as

$$\frac{\partial L_s^t}{\partial \boldsymbol{u}_z} = 2\kappa\frac{\partial \boldsymbol{l}_s^t}{\partial \boldsymbol{u}_z}r_s + 2\left\{ \begin{array}{l} \frac{\partial \boldsymbol{U}_s^t}{\partial \boldsymbol{u}_z}\mathrm{diag}(\bar{\boldsymbol{\gamma}}_s)[\widetilde{\boldsymbol{U}}_s^t - \bar{\boldsymbol{U}}]^{+} - \\ \frac{\partial \boldsymbol{U}_s^t}{\partial \boldsymbol{u}_z}\mathrm{diag}(\underline{\boldsymbol{\gamma}}_s)[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_s^t]^{+} + \\ \frac{\partial \boldsymbol{l}_s^t}{\partial \boldsymbol{u}_z}\mathrm{diag}(\boldsymbol{\gamma}_s)[\widetilde{\boldsymbol{l}}_s^t - \bar{\boldsymbol{l}}_s]^{+} \end{array} \right\} \tag{26}$$

By ignoring the sensitivities of the currents on the lines with respect to the injections at the nodes that are at the lines' upstream position, one gets $\partial l_{s.ij}^t/\partial \boldsymbol{u}_{z.\mathrm{in}} = 0 (s \in \mathcal{Z}^+,\ z \in \mathcal{Z}^+,\ z \neq s)$ and thus has

$$\frac{\partial L_s^t}{\partial \boldsymbol{u}_z} = 2\left\{ \begin{array}{l} \frac{\partial \boldsymbol{U}_s^t}{\partial \boldsymbol{u}_z}\mathrm{diag}(\bar{\boldsymbol{\gamma}}_s)[\widetilde{\boldsymbol{U}}_s^t - \bar{\boldsymbol{U}}]^{+} - \\ \frac{\partial \boldsymbol{U}_s^t}{\partial \boldsymbol{u}_z}\mathrm{diag}(\underline{\boldsymbol{\gamma}}_s)[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_s^t]^{+} \end{array} \right\}. \tag{27}$$

Based on the linearized power flow Eqs. (24), one can deduce that

$$\frac{\partial \boldsymbol{U}_s^t}{\partial \boldsymbol{u}_z} = \frac{\partial \boldsymbol{u}_{z.\text{in}}^t}{\partial \boldsymbol{u}_z}\frac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\text{in}}}\frac{\partial \boldsymbol{U}_s^t}{\partial U_{s.0}^t}, \tag{28}$$

and then transmit (28) into (27):

$$\frac{\partial L_s^t}{\partial \boldsymbol{u}_z} = 2\begin{Bmatrix} \frac{\partial \boldsymbol{u}_{z.\text{in}}^t}{\partial \boldsymbol{u}_z}\frac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\text{in}}}\frac{\partial \boldsymbol{U}_s^t}{\partial U_{s.0}^t}\text{diag}(\bar{\boldsymbol{\gamma}}_s)\Big)[\widetilde{\boldsymbol{U}}_s^t - \bar{\boldsymbol{U}}]^+ - \\ \frac{\partial \boldsymbol{u}_{z.\text{in}}^t}{\partial \boldsymbol{u}_z}\frac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\text{in}}}\frac{\partial \boldsymbol{U}_s^t}{\partial U_{s.0}^t}\text{diag}(\underline{\boldsymbol{\gamma}}_s)\Big)[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_s^t]^+ \end{Bmatrix}, \tag{29}$$

then

$$\begin{aligned} \frac{\partial L_s^t}{\partial \boldsymbol{u}_z} &= 2\frac{\partial \boldsymbol{u}_{z.\text{in}}^t}{\partial \boldsymbol{u}_z}\frac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\text{in}}}\begin{Bmatrix} \frac{\partial \boldsymbol{U}_s^t}{\partial U_{s.0}^t}\text{diag}(\bar{\boldsymbol{\gamma}}_s)[\widetilde{\boldsymbol{U}}_s^t - \bar{\boldsymbol{U}}]^+ - \\ \frac{\partial \boldsymbol{U}_s^t}{\partial U_{s.0}^t}\text{diag}(\underline{\boldsymbol{\gamma}}_s)[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_s^t]^+ \end{Bmatrix} \\ &= \frac{\partial \boldsymbol{u}_{z.\text{in}}^t}{\partial \boldsymbol{u}_z}\frac{\partial U_{s.0}^t}{\partial \boldsymbol{u}_{z.\text{in}}}\frac{\partial L_s^t}{\partial U_{s.0}^t}. \end{aligned} \tag{30}$$

Thus, the decoupled result of $\partial L_s^t/\partial \boldsymbol{u}_z$ has been deduced. One can also deduce the other terms in (8) and (10) in similar ways. It can be realized by decoupling $\partial U_{s.i}^t/\partial \boldsymbol{u}_{z.\text{in}}$ based on the interaction laws among different areas and submitting the decoupled results of $\partial U_{s.i}^t/\partial \boldsymbol{u}_{z.\text{in}}$ into the expanded versions of the corresponding terms (similar to (27)).

Appendix C. *Calculation of* (11) and (13)

The calculation of (11) and (13) is also based on the linearized power flow Eqs. (24).

a. For $\partial L_z^t/\partial \boldsymbol{u}_z$ and $\partial^2 L_z^t/\partial \boldsymbol{u}_z^2$ ($z \in \mathcal{Z}$),

$$\begin{cases} \frac{\partial L_z^t}{\partial \boldsymbol{u}_z} = \kappa\frac{\partial \boldsymbol{l}_z^t}{\partial \boldsymbol{u}_z}r_z + \partial f_z^t/\partial \boldsymbol{u}_z + 2\begin{Bmatrix} \frac{\partial \boldsymbol{U}_z^t}{\partial \boldsymbol{u}_z}\text{diag}(\bar{\boldsymbol{\gamma}}_z)\Big)[\widetilde{\boldsymbol{U}}_z^t - \bar{\boldsymbol{U}}]^+ - \\ \frac{\partial \boldsymbol{U}_z^t}{\partial \boldsymbol{u}_z}\text{diag}(\underline{\boldsymbol{\gamma}}_z)\Big)[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_z^t]^+ + \\ \frac{\partial \boldsymbol{l}_z^t}{\partial \boldsymbol{u}_z}\text{diag}(\boldsymbol{\gamma}_z)\Big)[\widetilde{\boldsymbol{l}}_z^t - \bar{\boldsymbol{l}}_z]^+ \end{Bmatrix} \\ \frac{\partial^2 L_z^t}{\partial \boldsymbol{u}_z^2} = \partial f_z^{t2}/\partial \boldsymbol{u}_z^2 + 2\begin{Bmatrix} \frac{\partial \boldsymbol{U}_z^t}{\partial \boldsymbol{u}_z}\text{diag}(\bar{\boldsymbol{\gamma}}_z \otimes \bar{\boldsymbol{I}}_z + \underline{\boldsymbol{\gamma}}_z \otimes \underline{\boldsymbol{I}}_z)\Big)\Big[\frac{\partial \boldsymbol{U}_z^t}{\partial \boldsymbol{u}_z}\Big]^{\mathrm{T}} + \\ \frac{\partial \boldsymbol{l}_z^t}{\partial \boldsymbol{u}_z}\text{diag}(\boldsymbol{\gamma}_z \otimes \boldsymbol{I}_z)\Big)\Big[\frac{\partial \boldsymbol{l}_z^t}{\partial \boldsymbol{u}_z}\Big]^{\mathrm{T}} \end{Bmatrix} \end{cases} \tag{31}$$

where $\bar{\boldsymbol{I}}_z$ and $\underline{\boldsymbol{I}}_z$ and $\boldsymbol{I}_z$ are the column vectors that have the same size as $\bar{\boldsymbol{\gamma}}_z$ and $\underline{\boldsymbol{\gamma}}_z$ and $\boldsymbol{\gamma}_z$ respectively, and the elements of these vectors are 1 if the corresponding power flow states exceed the limits or are 0.

b. For $\partial L_z^t/\partial U_{z.0}$ and $\partial^2 L_z^t/\partial U_{z.0}^2$ ($z \in \mathcal{Z}^+$),

$$\begin{cases} \frac{\partial L_z^t}{\partial U_{z.0}} = 2\{\bar{\boldsymbol{\gamma}}_z^{\mathrm{T}}[\widetilde{\boldsymbol{U}}_z^t - \bar{\boldsymbol{U}}]^+ - \underline{\boldsymbol{\gamma}}_z^{\mathrm{T}}[\underline{\boldsymbol{U}} - \widetilde{\boldsymbol{U}}_z^t]^+\} \\ \frac{\partial^2 L_z^t}{\partial U_{z.0}^2} = 2[\bar{\boldsymbol{\gamma}}_z \otimes \bar{\boldsymbol{I}}_z]^{\mathrm{T}}[\bar{\boldsymbol{\gamma}}_z \otimes \bar{\boldsymbol{I}}_z] + 2[\underline{\boldsymbol{\gamma}}_z \otimes \underline{\boldsymbol{I}}_z]^{\mathrm{T}}[\underline{\boldsymbol{\gamma}}_z \otimes \underline{\boldsymbol{I}}_z] \end{cases} \tag{32}$$

c. For $\partial^2 L_z^t/\partial \boldsymbol{u}_z \partial U_{z.0}$ ($z \in \mathcal{Z}^+$),

$$\frac{\partial^2 L_z^t}{\partial \boldsymbol{u}_z \partial U_{z.0}} = 2\frac{\partial \boldsymbol{U}_z^t}{\partial \boldsymbol{u}_z}\Big[\bar{\boldsymbol{\gamma}}_z \otimes \bar{\boldsymbol{I}}_z + \underline{\boldsymbol{\gamma}}_z \otimes \underline{\boldsymbol{I}}_z\Big] \tag{33}$$

d. For $\partial L_0^t/\partial \boldsymbol{u}_{z.\text{in}}$ and $\partial^2 L_0^t/\partial \boldsymbol{u}_{z.\text{in}}^2$ and $\partial^2 L_0^t/\partial \boldsymbol{u}_{z.\text{in}}\partial \boldsymbol{u}_{s.\text{in}}$ and $\partial^2 L_0^t/\partial \boldsymbol{u}_{z.\text{in}}\partial \boldsymbol{u}_0$ ($z \in \mathcal{Z}^+$, $s \in \mathcal{Z}^+$, $z \neq s$), they can be calculated in a similar way as (31), where $\boldsymbol{u}_{z.\text{in}}$ is considered as the output of a DER connected to the boundary node.

Next, we deal with the calculation of $\partial \boldsymbol{U}_z^t/\partial \boldsymbol{u}_z$, $\partial \boldsymbol{l}_z^t/\partial \boldsymbol{u}_z$ in (31) and $\partial U_{s.0}^t/\partial \boldsymbol{u}_{z.0}$, $\partial \boldsymbol{u}_{z.\text{in}}^t/\partial \boldsymbol{u}_z$, $\partial U_{s.0}^t/\partial \boldsymbol{u}_{z.\text{in}}$ in (11), (13). These terms are or can be equivalent to the matrixes composed of the sensitivities of power flow states with respect to DER outputs. These sensitivities can be calculated as

$$\begin{cases} \frac{\partial P_{z.ij}^t}{\partial P_{z.k}} = -\prod_{mn \in \mathcal{M}_{z.i}^{\text{dn}} \cap \mathcal{M}_{z.k}^{\text{up}}} \frac{1}{1 - 2\widetilde{P}_{z.mn}^t r_{z.mn}/\widetilde{U}_{z.i}^t} \\ \frac{\partial Q_{z.ij}^t}{\partial Q_{z.k}} = -\prod_{mn \in \mathcal{M}_{z.i}^{\text{dn}} \cap \mathcal{M}_{z.k}^{\text{up}}} \frac{1}{1 - 2\widetilde{Q}_{z.mn}^t x_{z.mn}/\widetilde{U}_{z.i}^t} \end{cases} \tag{34a}$$

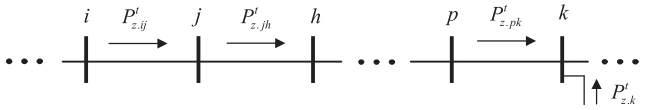


Fig. 15. Simplified graph for distribution grids.

$$\frac{\partial l^t_{z.ij}}{\partial P_{z.k}} = -\frac{2\widetilde{P}^t_{z.ij}}{\widetilde{U}^t_{z.i}}\frac{\partial P^t_{z.ij}}{\partial P_{z.k}},\ \ \frac{\partial l^t_{z.ij}}{\partial Q_{z.k}} = -\frac{2\widetilde{Q}^t_{z.ij}}{\widetilde{U}^t_{z.i}}\frac{\partial Q^t_{z.ij}}{\partial Q_{z.k}} \tag{34b}$$

$$\begin{cases} \frac{\partial U^t_{z.i}}{\partial P_{z.k}} = \frac{\partial U^t_{z.0}}{\partial P_{z.\mathrm{up}}}\frac{\partial P^t_{z.\mathrm{up}}}{\partial P_{z.k}} + \\ 2\sum_{mn\in\mathcal{M}^{\mathrm{up}}_{z.i}\cap\mathcal{M}^{\mathrm{up}}_{z.k}}\left[(\widetilde{P}^t_{z.mn}|z_{z.mn}|^2/\widetilde{U}^t_{z.i} - r_{z.mn})\frac{\partial P^t_{z.mn}}{\partial P_{z.k}}\right] \\ \frac{\partial U^t_{z.i}}{\partial Q_{z.k}} = \frac{\partial U^t_{z.0}}{\partial Q_{z.\mathrm{up}}}\frac{\partial Q^t_{z.\mathrm{up}}}{\partial Q_{z.k}} + \\ 2\sum_{mn\in\mathcal{M}^{\mathrm{up}}_{z.i}\cap\mathcal{M}^{\mathrm{up}}_{z.k}}\left[(\widetilde{Q}^t_{z.mn}|z_{z.mn}|^2/\widetilde{U}^t_{z.i} - x_{z.mn})\frac{\partial Q^t_{z.mn}}{\partial Q_{z.k}}\right] \end{cases} \tag{34c}$$

where $\mathcal{M}^{\mathrm{up}}_{z.i}$, $\mathcal{M}^{\mathrm{dn}}_{z.i}$ are the sets of nodes on the upstream and downstream of node i respectively.

(34) is deduced as follows. We give a visualization as Fig. 15 to understand the deduction.

For $\partial P^t_{z.ij}/\partial P_{z.k}$ in (34c), by utilizing the derivative chain rule and neglecting the interaction sensitivities between real powers and reactive powers, one can obtain:

$$\frac{\partial P^t_{z.ij}}{\partial P_{z.k}} = \frac{\partial P^t_{z.ij}}{\partial P_{z.jh}}\cdots\frac{\partial P^t_{z.pk}}{\partial P_{z.k}} \tag{35}$$

and since

$$\frac{\partial P^t_{z.ij}}{\partial P_{z.jh}} = \frac{\partial P^t_{z.ij}}{\partial P_{z.j}} \tag{36}$$

one has

$$\frac{\partial P^t_{z.ij}}{\partial P_{z.k}} = \frac{\partial P^t_{z.ij}}{\partial P_{z.j}}\cdots\frac{\partial P^t_{z.pk}}{\partial P_{z.k}} \tag{37}$$

where

$$\frac{\partial P^t_{z.ij}}{\partial P^t_{z.j}} = \frac{1}{2r_{ij}\widetilde{P}^t_{z.ij}/\widetilde{U}^t_{z.i}-1} \tag{38}$$

(38) is obtained by taking the derivative of (24b) with respect to $P_{z.j}$ and neglecting the interaction sensitivities between real powers and reactive powers.

Submitting (38) into (37), one can obtain $\partial P^t_{z.ij}/\partial P_{z.k}$. In similar ways, one can obtain $\partial Q^t_{z.ij}/\partial Q_{z.k}$.

For (34b), one can deduce it by taking the derivative of (24d) with respect to $P_{z.j}$ and $Q_{z.j}$ respectively.

For (34c), by accumulating (24a) from the root node to node i, one can get

$$U^t_{z.i} = U^t_{z.0} - \sum_{mn\in\mathcal{M}^{\mathrm{up}}_{z.(i,k)}}[2(r_{z.ij}P^t_{z.ij} + x_{z.ij}Q^t_{z.ij}) + |z_{ij}|^2\, h^t_{ij}(P^t_{z.ij}, Q^t_{z.ij})], \tag{39}$$

and then by taking the derivative of (39) with respect to $P_{z.j}$ and $Q_{z.j}$ respectively, one can obtain (34c).

Appendix D. Strategy to adjust the optimizing step length

We take the adjustment for $\Delta P^t_{z.i}$ as an example. Set $\Delta U^t_{\max} = \mathrm{Max}\{\widetilde{\boldsymbol{U}}^t_z - \bar{\boldsymbol{U}}\}_{z\in\mathcal{Z}}$, $\Delta U^t_{\min} = \mathrm{Min}\{\widetilde{\boldsymbol{U}}^t_z - \underline{\boldsymbol{U}}\}_{z\in\mathcal{Z}}$, $\Delta l^t_{\max} = \mathrm{Max}\{\widetilde{\boldsymbol{l}}^t_z - \bar{\boldsymbol{l}}_z\}_{z\in\mathcal{Z}}$, $\Delta P^t_{\mathrm{loss}} = P^t_{\mathrm{loss}} - P^{t-1}_{\mathrm{loss}}$; set $\Delta P_{\max}$, $\Delta P_{\min}$ as the maximum and minimum values of the step length; set ε_{U}, ζ_{U}, ε_l, $\varepsilon_{\mathrm{loss}}$ as the threshold-related values to trigger the adjust strategy; set $\Delta P^t_{z.i.1}$, $\Delta P^t_{z.i.2}$, $\Delta P^t_{z.i.3}$, $\Delta P^t_{z.i.4}$ as the copies of $\Delta P^t_{z.i}$ that take as $\Delta P_{\max}$ except the situations as follows.

For $i\in\mathcal{N}_z$, $ij\in\mathcal{M}_z$, $z\in\mathcal{Z}$,

(1) If $-\zeta_{\mathrm{U}} \leqslant \Delta U^t_{\max} \leqslant \varepsilon_{\mathrm{U}}$,

$$\Delta P^t_{z.i.1} = a^t_{\mathrm{U}}(\Delta U^t_{\max})^2 + \Delta P_{\min} \tag{40a}$$

where

$$\begin{cases} a^t_{\mathrm{U}} = (\Delta P_{\max} - \Delta P_{\min})/(\varepsilon_{\mathrm{U}})^2,\ 0< \Delta U^t_{\max} \leqslant \varepsilon_{\mathrm{U}} \\ a^t_{\mathrm{U}} = (\Delta P_{\max} - \Delta P_{\min})/(\zeta_{\mathrm{U}})^2,\ -\zeta_{\mathrm{U}} \leqslant \Delta U^t_{\max} \leqslant 0 \end{cases}. \tag{40b}$$

(2) If $-\varepsilon_{\mathrm{U}} \leqslant \Delta U^t_{\min} \leqslant \zeta_{\mathrm{U}}$,

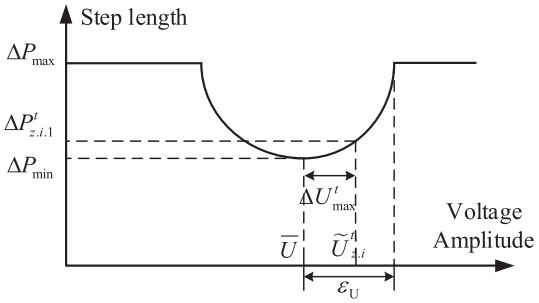


Fig. 16. Visualization of the adjustment for the step length.

$$\Delta P^t_{z.i.2} = a^t_U(\Delta U^t_{\min})^2 + \Delta P_{\min} \tag{41a}$$

where

$$\begin{cases} a^t_U = (\Delta P_{\max} - \Delta P_{\min})/(\varepsilon_U)^2, -\varepsilon_U \leqslant \Delta U^t_{\min} < 0 \\ a^t_U = (\Delta P_{\max} - \Delta P_{\min})/(\zeta_U)^2, 0 \leqslant \Delta U^t_{\min} \leqslant \zeta_U \end{cases}. \tag{41b}$$

(3) If $-\varepsilon_l \leqslant \Delta l^t_{\max} \leqslant \varepsilon_l$,

$$\Delta P^t_{z.i.3} = a^t_l(\Delta l^t_{\max})^2 + \Delta P_{\min} \tag{42a}$$

where

$$a^t_U = (\Delta P_{\max} - \Delta P_{\min})/(\varepsilon_l)^2. \tag{42b}$$

(4) If $\Delta P_{\text{loss.0}} - \varepsilon_{\text{loss}} \leqslant \Delta P^t_{\text{loss}} \leqslant \Delta P_{\text{loss.0}} + \varepsilon_{\text{loss}}$,

$$\Delta P^t_{z.i.4} = a^t_{\text{loss}}(\Delta P^t_{\text{loss}} - \Delta P_{\text{loss.0}})^2 + \Delta P_{\min} \tag{43a}$$

where

$$a^t_{\text{loss}} = (\Delta P_{\max} - \Delta P_{\min})/(\varepsilon_{\text{loss}})^2. \tag{43b}$$

Then, we take the step length as

$$\Delta P^t_{z.i} = \text{Min}\{\Delta P^t_{z.i.1}, \Delta P^t_{z.i.2}, \Delta P^t_{z.i.3}, \Delta P^t_{z.i.4}\} \tag{44}$$

The above strategy can decrease the step length when the voltages or currents are adjacent to the boundaries, or when the objective is adjacent to the optimal value. This adjusting process can be visualized as Fig. 16. One can see that the step length will be decreased as quadratic properties when the voltage is adjacent to its upper boundary.

Noted that the adjusting strategy can be performed in a distributed way as follows. Firstly, each area calculates the variables about the operation states only related to itself as $\Delta U^t_{\max.z} = \text{Max}\{\widetilde{U}^t_z - \bar{U}\}$. Then the upper area collects the calculating results together to furtherly calculate the related variables as $\Delta U^t_{\max} = \text{Max}\{\Delta U^t_{\max.z}\}_{z\in\mathcal{Z}}$. Finally, the upper area adjusts the step length based on (40)–(44), and sends the results to each area to implement the adjustment.

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